

Exact differentiation of the AVC update law, and why control requires a hybrid gradient

SPECULA AVC study — `avc_full_demo`

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Abstract

The AVC algorithm of Muradore et al. (SPIE 8447-38) updates its state along the regressor $\mathbf{W}$, treating $\mathbf{W}$ as independent of the state even though it depends on it through the control parameters. We derive the exact gradient in closed form and show that it is useless as a control law: with the published regularisation ($\epsilon = 0$) it vanishes *identically*, because the cost being minimised does not depend on the parameters being optimised. In particular the exact gradient gives the disturbance quadratures no drive, so their estimate remains at zero and no cancellation occurs. The pseudo-gradient is not an approximation to the exact gradient but a different object—integral action on the physical residual—which we show converges to *exact* tone cancellation whenever the plant estimate’s phase error is below 90° . This motivates a hybrid update: pseudo-gradient on the disturbance quadratures, exact gradient on the plant parameters. In a controlled closed-loop simulation the hybrid cancels a 47 Hz vibration by 48–59 dB, whereas the original update provides no cancellation at all.

1 Setup

One AVC instance tracks a sinusoidal disturbance of phase α . Its state holds the real and imaginary parts of the plant frequency response and of the disturbance quadratures,

$$\mathbf{x} = (x_1, x_2, x_3, x_4)^\top, \quad P = x_1 + ix_2, \quad \Lambda = x_3 + ix_4. \quad (1)$$

From $\mathbf{x}$ the algorithm forms the control quadratures $\vartheta = \theta_c + i\theta_s$ (their eq. 11),

$$\theta_c = -\frac{x_1x_3 - x_2x_4}{D}, \quad \theta_s = -\frac{x_1x_4 + x_2x_3}{D}, \quad D = x_1^2 + x_2^2 + \epsilon \quad \Longleftrightarrow \quad \vartheta = -\frac{P\Lambda}{D}, \quad (2)$$

the equivalence following from $P\Lambda = (x_1x_3 - x_2x_4) + i(x_1x_4 + x_2x_3)$. Here $\epsilon \geq 0$ regularises the division: the denominator is $|P|^2$, the squared magnitude of the *plant estimate*, so the control gain diverges if the estimate approaches the origin. The paper takes $\epsilon = 0$; ESO’s `updateAVC.m` guards the division with a discontinuous clamp; we use the smooth equivalent with $\epsilon = \theta_{\min} = 10^{-2}$ by default. With $c \equiv \cos \alpha$, $s \equiv \sin \alpha$ the regressor (their eq. 9) and the update (their eq. 10, step $\mu = g_x T_s$) are

$$\mathbf{W} = (\theta_c c + \theta_s s, \theta_s c - \theta_c s, c, s)^\top, \quad \hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} - \mu \mathbf{W} e, \quad e = \mathbf{W}^\top \hat{\mathbf{x}} - y, \quad (3)$$

with $J = \frac{1}{2}e^2$ the cost and y the measured residual.

2 Exact differentiation

Since $\mathbf{W} = \mathbf{W}(\vartheta(\hat{\mathbf{x}}))$ depends on the estimate, the derivative of the error is not $\mathbf{W}$:

$$\frac{\partial e}{\partial x_i} = W_i + \sum_j \frac{\partial W_j}{\partial x_i} x_j = (\mathbf{W} + J\mathbf{W}\mathbf{x})_i \equiv g_i, \quad (J\mathbf{W})_{ji} = \frac{\partial W_j}{\partial x_i}. \quad (4)$$

The paper drops the second term. We compute it exactly.

Step 1: gradients of the control parameters. Writing $\theta_c = N_c/D$ with $N_c = -(x_1x_3 - x_2x_4)$ and applying the quotient rule in the form $\partial_i\theta_c = \partial_iN_c/D - \theta_c\partial_iD/D$ (and likewise for θ_s), with $\nabla D = (2x_1, 2x_2, 0, 0)$,

$$\nabla\theta_c = \frac{1}{D} \begin{pmatrix} -x_3 - 2x_1\theta_c \\ x_4 - 2x_2\theta_c \\ -x_1 \\ x_2 \end{pmatrix}, \quad \nabla\theta_s = \frac{1}{D} \begin{pmatrix} -x_4 - 2x_1\theta_s \\ -x_3 - 2x_2\theta_s \\ -x_2 \\ -x_1 \end{pmatrix}. \quad (5)$$

Step 2: the chain-rule term. Only W_1, W_2 depend on $\mathbf{x}$, through ϑ , and $\partial_iW_1 = c\partial_i\theta_c + s\partial_i\theta_s$, $\partial_iW_2 = c\partial_i\theta_s - s\partial_i\theta_c$. Hence

$$(J_{\mathbf{W}\mathbf{x}}^\top)_i = x_1\partial_iW_1 + x_2\partial_iW_2 = A\partial_i\theta_c + B\partial_i\theta_s, \quad \begin{aligned} A &= x_1c - x_2s, \\ B &= x_1s + x_2c, \end{aligned} \quad (6)$$

so that $A + iB = Pe^{i\alpha}$.

Step 3: the disturbance components. Using $\partial_3\theta_c = -x_1/D$, $\partial_3\theta_s = -x_2/D$ in (6) and the identities $Ax_1 + Bx_2 = c(x_1^2 + x_2^2)$, $Ax_2 - Bx_1 = -s(x_1^2 + x_2^2)$,

$$(J_{\mathbf{W}\mathbf{x}}^\top)_3 = -\frac{x_1^2 + x_2^2}{D}c \quad \implies \quad \boxed{g_3 = c\frac{\epsilon}{D}, \quad g_4 = s\frac{\epsilon}{D}} \quad (7)$$

since $D - (x_1^2 + x_2^2) = \epsilon$.

Step 4: the plant components. Carrying (5)–(6) through in complex form, using $\vartheta = -P\Lambda/D$ and $\Lambda e^{-i\alpha} + \bar{\Lambda}e^{i\alpha} = 2\mathcal{R}$ with $\mathcal{R} \equiv \text{Re}(\Lambda e^{i\alpha})$,

$$\boxed{g_1 + ig_2 = -2\mathcal{R}\frac{\epsilon}{D^2}P} \quad (8)$$

a *real* multiple of P : the exact plant update can only rescale the plant estimate, never rotate it.

All of (5)–(8) agree with a finite-difference Jacobian to $\sim 10^{-9}$ relative error.

3 The cost is not a control objective

Both boxed results carry a factor ϵ . This is not a coincidence.

The model predicts perfect cancellation of its own estimate. Collecting the c and s parts of $\mathbf{W}^\top\mathbf{x}$ in complex notation gives $\mathbf{W}^\top\mathbf{x} = \text{Re}(Z)\cos\alpha + \text{Im}(Z)\sin\alpha$ with $Z = \vartheta\bar{P} + \Lambda$. Substituting the control (2), which is built from the estimate itself,

$$\hat{Z} = -\frac{\hat{P}\hat{\Lambda}}{D}\bar{\hat{P}} + \hat{\Lambda} = \hat{\Lambda}\left(1 - \frac{|\hat{P}|^2}{D}\right) = \hat{\Lambda}\frac{\epsilon}{D} \xrightarrow{\epsilon \rightarrow 0} 0. \quad (9)$$

Consequence: at $\epsilon = 0$ the exact gradient is identically zero.

$$\epsilon = 0 \implies \mathbf{W}^\top\hat{\mathbf{x}} \equiv 0 \quad \forall \hat{\mathbf{x}} \implies J = \frac{1}{2}(\mathbf{W}^\top\hat{\mathbf{x}} - y)^2 = \frac{1}{2}y^2 \implies \mathbf{g} \equiv \mathbf{0}. \quad (10)$$

Because ϑ is constructed to cancel $\hat{\Lambda}$, the model always predicts perfect cancellation of its own estimate—whatever that estimate is. The cost therefore *does not depend on the parameters*

being optimised, and a cost independent of its arguments has zero gradient. Numerically, at $\epsilon = 0$ we measure $\|\mathbf{g}\| \sim 4 \times 10^{-15}$ (machine zero) against $\|\mathbf{W}\| \sim 20\text{--}40$.

This is the central point. The quantity $e = \mathbf{W}^\top \hat{\mathbf{x}} - y$ is an *equation error*—a measure of model self-consistency—not a control error. Minimising it exactly achieves nothing, because the model is self-consistent by construction. What one actually wants to minimise is $|y|$, the physical residual. The two objectives are different, and the published update happens to serve the second while purporting to serve the first.

Practical consequence: the disturbance estimate collapses. By (7) the exact drive on x_3, x_4 is the pseudo-gradient attenuated by $\epsilon/D \approx \epsilon/|P|^2$ — in the simulation of Sec. 6, $\sim 2 \times 10^{-4}$. Started from $\hat{\Lambda} = 0$, the disturbance estimate therefore stays at zero, no correction is generated, and the residual is that of the uncontrolled loop. This is exactly what is observed when the exact gradient is applied to all four components in the closed loop of Sec. 6: the plant estimate is pinned at its seed, the disturbance estimate never leaves the origin ($|\hat{\Lambda}| \rightarrow 0$), and the 47 Hz residual is that of the baseline.

4 What the pseudo-gradient does, and why to keep it

Since $\mathbf{W}^\top \hat{\mathbf{x}} = \mathcal{O}(\epsilon)$ by (9), the error reduces to $e \simeq -y$ and the disturbance part of (3) becomes, in complex form,

$$\hat{\Lambda} \leftarrow \hat{\Lambda} + \mu y e^{i\alpha}. \quad (11)$$

Averaging over one cycle with $y = \text{Re}(Z) \cos \alpha + \text{Im}(Z) \sin \alpha$ and $\langle c^2 \rangle = \langle s^2 \rangle = \frac{1}{2}$, $\langle cs \rangle = 0$ gives $\langle y e^{i\alpha} \rangle = \frac{1}{2}Z$, i.e. the slow dynamics

$$\dot{\hat{\Lambda}} \propto Z = \Lambda_\star - \kappa \hat{\Lambda}, \quad \kappa = \overline{\left(\frac{P_\star}{\hat{P}} \right)}, \quad (12)$$

where $P_\star, \Lambda_\star$ are the true plant and disturbance and Z is the residual phasor actually measured. Two conclusions follow immediately.

- **The fixed point is exact cancellation.** At $\hat{\Lambda}_\infty = \Lambda_\star/\kappa$ one has $Z = 0$: the tone is removed *completely*, for any plant estimate. A magnitude error in $\hat{P}$ merely rescales $\hat{\Lambda}_\infty$ and changes the convergence rate; it does not degrade the steady state.
- **Stability requires only a phase margin.** Linearising (12), the error decays as $e^{-\kappa t}$, so the loop is stable iff $\text{Re } \kappa > 0$, i.e.

$$|\arg P_\star - \arg \hat{P}| < 90^\circ. \quad (13)$$

Table 1 verifies both statements in an idealised setting (no delay, no noise, plant estimate frozen with a deliberate phase error).

The pseudo-gradient is thus a certainty-equivalence integral controller on the physical residual, and it is the component that makes the algorithm work. It must be retained.

5 The hybrid update

Sections 3–4 assign each half of the state its appropriate objective:

$$\hat{x}_{1,2} \leftarrow \hat{x}_{1,2} - \mu c_p g_{1,2} e \quad (\text{exact}), \quad \hat{x}_{3,4} \leftarrow \hat{x}_{3,4} - \mu W_{3,4} e \quad (\text{pseudo / integral}). \quad (14)$$

The disturbance keeps the integral action of Sec. 4. The plant receives the exact direction (8), which by construction is *radial*: it rescales $\hat{P}$ without rotating it, so it cannot destroy the phase margin (13) on which cancellation depends. Under the pseudo-gradient, by contrast, the plant update has a large tangential component, $\arg \hat{P}$ wanders freely, the margin is violated and the loop diverges.

phase error	final $ Z $ (residual tone)	final $ \hat{\Lambda} $	
0°	5.7×10^{-11}	600.00	converged
20°	2.1×10^{-11}	600.00	converged
45°	3.3×10^{-11}	600.00	converged
70°	7.3×10^{-7}	600.00	converged
85°	3.2	603	marginal
91°	1.7×10^3	2296	unstable
120°	∞	∞	diverged

Table 1: Verification of (12)–(13). True $|\Lambda_\star| = 600$. Below 90° the tone is cancelled to numerical precision regardless of the plant estimate; beyond 90° the loop diverges.

Damping. The plant drive (8) has magnitude $2|\mathcal{R}|\epsilon|P|/D^2$, which is small while $|P|^2 \gg \epsilon$ but grows as $|P|$ shrinks; since the update is radial, a shrinking $|\hat{P}|$ raises its own gain. The constant c_p (the c parameter) damps this state-dependent runaway. Empirically, in the closed loop of Sec. 6, $c_p = 1$ diverges while $c_p = 0.1$ settles; $c_p \lesssim 0.1$ is used throughout.

Scope. The hybrid does not *identify* the plant: by (8) the phase is a conserved quantity of the update, and $|\hat{P}|$ settles at a value that need not equal $|P_\star|$. It provides a bounded, phase-preserving magnitude adjustment around a seeded operating point. The plant phase must still be supplied by calibration, within the 90° margin.

6 Simulation results

Closed-loop single-conjugate AO simulation in SPECULA: 64 px pupil, 8×8 Shack–Hartmann, 40 Zernike modes, 1 kHz, integrator (gain 0.5, 2-frame delay) on all modes, first-order 80 Hz actuator low-pass, and a 120 nm tip vibration at 47 Hz. AVC acts on mode 0 in parallel with the integrator. At 47 Hz the loop *amplifies*: the injected tone emerges as a 891 nm residual ($\times 7.4$), which is what makes a narrow-band canceller worthwhile.

Six 10 s runs share the atmospheric seed and vibration realisation and differ *only* in the update rule and the plant seed; the seeds are the measured closed-loop value ($|P| = 7.22 \angle 169.5^\circ$) and a deliberately wrong one ($|P| = 1.41 \angle 135^\circ$, a 34.5° phase error, inside the margin (13)). Statistics are taken over the second half of each run.

update	seed	mode-0 RMS	all-mode RMS	47 Hz amp.	vs. base	final $ \hat{P} $
none (baseline)	—	630.4 nm	99.9 nm	891 nm	—	—
ORIGINAL pseudo	wrong	640.7 nm	101.5 nm	898 nm	+0.1 dB	205 (wandering)
ORIGINAL pseudo	correct	639.6 nm	101.3 nm	896 nm	+0.0 dB	64 (wandering)
HYBRID	wrong	36.5 nm	7.3 nm	3.5 nm	−48.2 dB	13.8 (settled)
HYBRID	correct	27.5 nm	6.2 nm	1.0 nm	−58.9 dB	4.0 (settled)
frozen plant	correct	26.5 nm	6.1 nm	0.6 nm	−63.0 dB	7.22 (fixed)

Table 2: Controlled comparison. The original update with online plant adaptation gives no cancellation, independently of how well it is initialised. The hybrid cancels by 48–59 dB from either seed. A measured, frozen plant is the best performer, consistent with Sec. 5: online plant adaptation buys robustness to a drifting plant, not accuracy.

The measurements match the theory point by point: the original update violates the phase margin and fails regardless of initialisation; the hybrid preserves it by construction and converges

Original (pseudo-gradient) vs corrected (exact-gradient hybrid) AVC
identical closed loop, 120 nm @ 47 Hz vibration, only the state-update differs

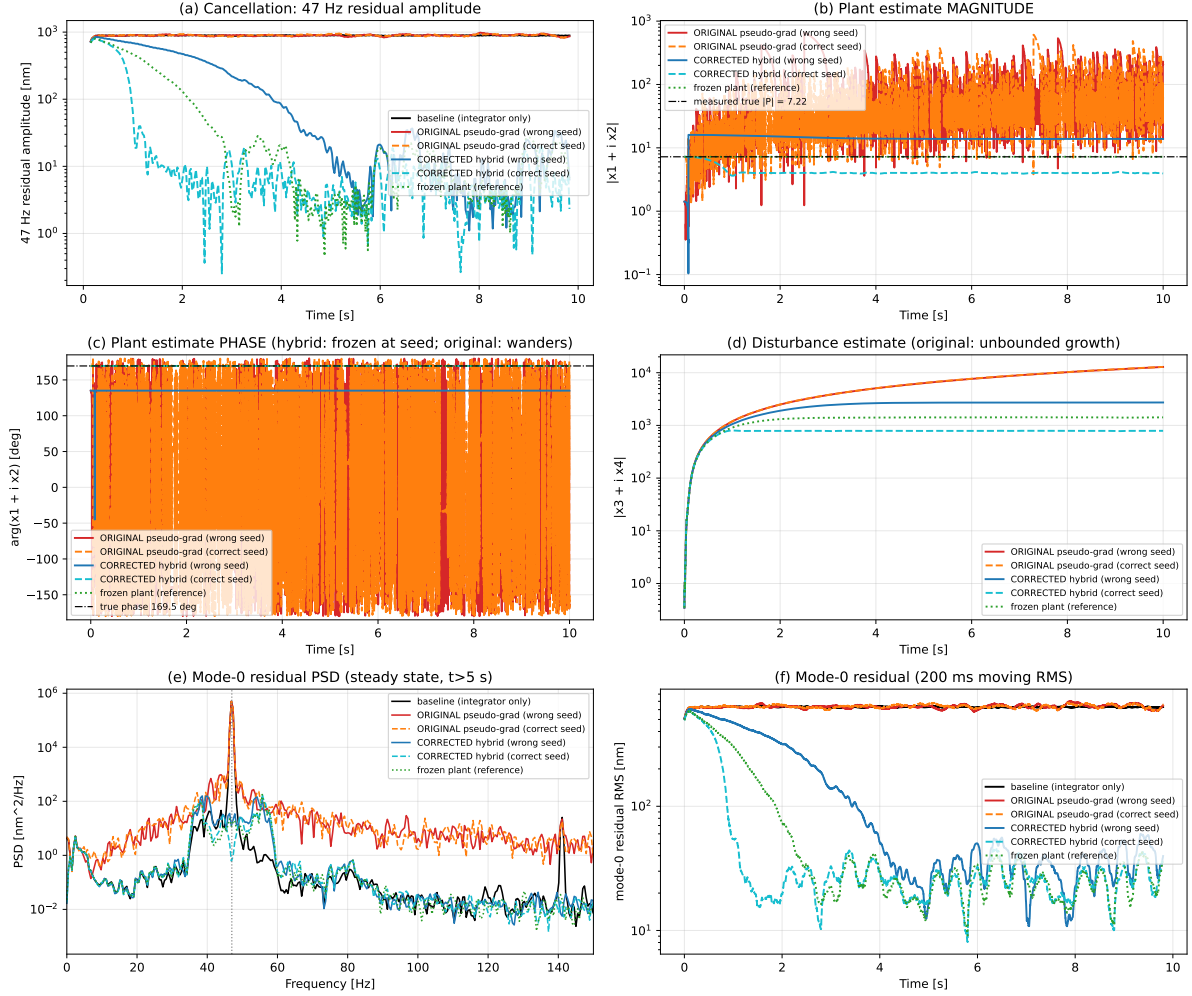


Figure 1: (a) the two ORIGINAL traces remain on the baseline while the hybrid and the frozen reference fall by 2–3 decades; (b),(c) the original plant estimate wanders over three decades in magnitude and the full $\pm 180^\circ$ in phase—violating (13)—whereas the hybrid settles and holds its seed phase exactly, as (8) requires; (d) the original disturbance estimate grows without bound while the hybrid plateaus; (e) in the steady-state PSD the original *enlarges* the 47 Hz peak; (f) mode-0 residual.

from either seed; the plant phase in the hybrid runs is numerically constant, as (8) predicts; and the wrong-seed run converges because its 34.5° error is inside the 90° margin.

7 Conclusion

The exact gradient of the AVC cost is not a usable control law: with the published regularisation it is identically zero, because the equation-error cost does not depend on the parameters being optimised. Restoring it in full removes the drive on the disturbance quadratures, whose estimate then stays at zero and cancels nothing. The pseudo-gradient, though not a gradient of the stated cost, is integral action on the physical residual and converges to exact tone cancellation under a 90° phase-margin condition on the plant estimate. Retaining it on the disturbance while applying the exact—and provably radial, hence margin-preserving—gradient to the plant yields an update that cancels a 47 Hz vibration by 48–59 dB where the published update cancels nothing.

Reproducibility. `avc_chainrule_check.py` (Jacobian and Eq. (7) against finite differences), `params_scao_avc_full_demo.yml` (update variant selected by `adapt_plant` and `exact_gradient`), `params_scao_vibration_baseline.yml`, `params_measure_plant.yml` (plant measurement by probe injection). A longer companion report, `avc_exact_gradient.pdf`, covers the observability analysis and the full experimental history.